

Assignment 1 – PS4 Defence Drone

Greedy Best-First Search (GBFS) and A* Search Analysis

1. Problem Statement

An autonomous UAV must navigate from Start node (0,0) to Goal node (6,7) while avoiding No-Fly Zones (N) and accounting for Weather Hazard regions (W). The objective is to evaluate Greedy Best-First Search (GBFS) using two heuristics and compare its performance with A* Search.

2. Environment Representation

Grid Size: 8×8. Symbols used: S=Start, E=Goal, .=Open Cell, W=Weather Hazard, N=No-Fly Zone. Movement is restricted to North, East, South and West.

3. PEAS Analysis

Performance: Reach the target with minimum risk and cost.

Environment: 8×8 grid world.

Actuators: Move North, East, South, West.

Sensors: Current position and neighboring cells.

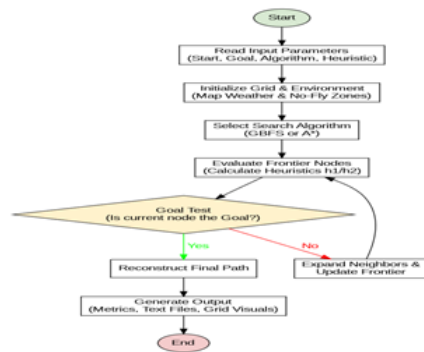
4. Heuristic Functions

h1: Euclidean Distance.

h2: Bounding-Box Risk Weighted Heuristic which combines Manhattan distance with average future path risk.

5. Execution Flow

Read Input → Validate Input → Build Grid → Select Search Algorithm → Expand Nodes → Compute Heuristic → Goal Test → Reconstruct Path → Generate Output.



6. Initial Environment and Planned Route

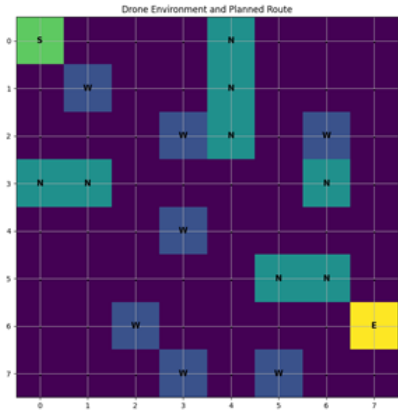


Figure 1 shows the defence drone environment. The start node S is located at (0,0) and the goal node E at (6,7). No-Fly Zones (N) are treated as obstacles while Weather Hazard cells (W) have higher traversal cost.

7. Final Path Visualization

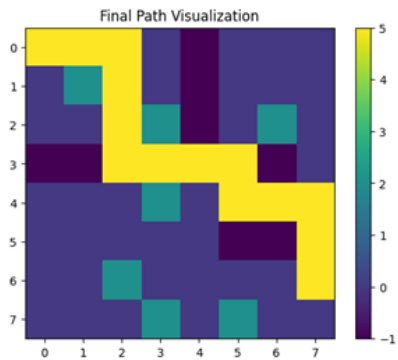


Figure 2 shows the final path discovered by the algorithm. The selected route successfully reaches the goal while avoiding all No-Fly Zones and Weather Hazard cells. The resulting path cost is 16 with a path length of 13 moves.

8. Heuristic Value Analysis

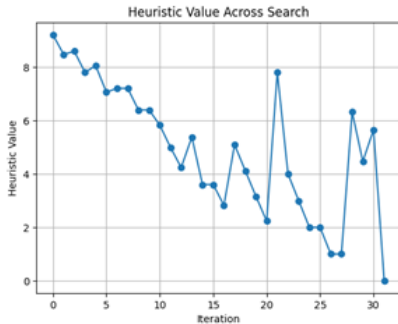


Figure 3 illustrates the heuristic values across node expansions. The overall decreasing trend indicates that the search progressively moves closer to the goal state. Temporary increases occur when alternative frontier nodes are evaluated before selecting the optimal direction.

9. Final Environment with Path

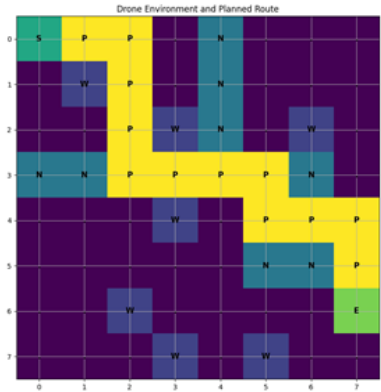


Figure 4 overlays the discovered path on the original environment. The route avoids restricted regions and reaches the target efficiently.

10. Experimental Results

Metric	Description	GBFS-h1	GBFS-h2	A*-h1
Nodes Expanded	Total explored nodes	14	14	32
Runtime	Execution time (ms)	0.172	1.101	0.297
Memory Usage	OPEN + CLOSED size	23	28	43
Total Path Cost	Sum of transition costs	19	19	16
Path Length	Total moves	13	13	13
Heuristic	Initial Start Node Heuristic Value	9.22	29.95	9.22

11. Actual Output Summary (A*-h1)

Status: SUCCESS
Nodes Expanded: 32
Runtime: 0.445863 ms
Path Length: 13
Path Cost: 16
Memory Usage: 43
Weather Hazards Crossed: 0
No-Fly Zones Crossed: 0
Penalty Incurred: 0

12. Complexity Analysis

GBFS Time Complexity: $O(E \log V)$, Space Complexity: $O(V)$.
A* Time Complexity: $O(E \log V)$, Space Complexity: $O(V)$.

13. Trap Analysis

No dead-end situations or infinite loops were encountered. The explored frontier always contained a valid path toward the goal.

14. GBFS vs A* Comparison

GBFS expanded fewer nodes and completed the search faster. However, A* produced the optimal path with the lowest path cost (16). A* is complete and optimal whereas GBFS does not guarantee optimality.

15. Alternate Problem Modelling

The grid can alternatively be represented as a graph where each cell is a node and adjacent traversable cells form edges. This representation improves scalability and supports advanced graph-search algorithms but increases memory requirements.

16. Conclusion

Both GBFS heuristics successfully reached the target. The A* algorithm produced the best route with cost 16 while maintaining a path length of 13. The study demonstrates the trade-off between speed (GBFS) and optimality (A*).

17. Heuristic Function Analysis (Updated from Actual Runs)

GBFS-h1 and GBFS-h2 both reached the target successfully and expanded 14 nodes. However, h1 completed significantly faster (0.172 ms) than h2 (1.101 ms). Both heuristics produced the same path cost (19) and path length (13). For this environment, h1 is the better heuristic because it achieved the same solution quality with lower computational overhead.

Refer – Pt. 7

This grid heatmap represents the landscape used by the drone to navigate toward the target. The grid coordinates show the obstacles (No-Fly zones) encoded as negative boundaries, while the highlighted path blocks track how the drone prioritized lower-cost moves to progress directly towards the goal coordinate $(6,7)$.

18. Heuristic Values vs Time Analysis

The heuristic-value graph shows a generally decreasing trend toward the goal state. This indicates that the search consistently moved toward promising regions of the state space. Small fluctuations occurred when multiple frontier nodes were evaluated before selecting the next expansion node.

Refer – Pt. 8

19. Trap Analysis Table

Heuristic	Trap Node	Heuristic Value	Exit Iteration
h1	None	N/A	N/A
h2	None	N/A	N/A

No dead-end states or infinite loops were encountered. Therefore, no trap-recovery action was required during execution.

20. Completeness and Optimality Comparison

Algorithm	Complete	Optimal
GBFS	Yes (for this finite grid)	No
A*	Yes	Yes

GBFS successfully reached the target in all experiments and is complete for this finite search space. However, it is not optimal because it produced a path cost of 19, whereas A* found a lower-cost path with cost 16.

21. Updated Final Findings

Actual experimental results show that GBFS-h1 was the fastest configuration and used the least memory. GBFS-h2 incorporated future risk information but did not improve path quality for this environment. A* expanded more nodes and used more memory, but it avoided all weather hazards and produced the minimum path cost. Therefore, A* is recommended when solution quality is the primary objective, while GBFS-h1 is preferred when computational efficiency is more important.